MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

07

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 A

A. $-12, 0$

Choice A is correct. It's given that $y = fx$. The x-intercepts of a graph in the xy-plane are the points where $y = 0$. Thus, for an x-intercept of the graph of function f, $0 = fx$. Substituting 0 for fx in the equation $fx = x^2 - 18x - 360$ yields $0 = x^2 - 18x - 360$. Factoring the right-hand side of this equation yields $0 = x + 12x - 30$. By the zero product property, $x + 12 = 0$ and $x - 30 = 0$. Subtracting 12 from both sides of the equation $x + 12 = 0$ yields $x = -12$. Adding 30 to both sides of the equation $x - 30 = 0$ yields $x = 30$. Therefore, the x-intercepts of the graph of $y = fx$ are $-12, 0$ and $30, 0$. Of these two x-intercepts, only $-12, 0$ is given as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2 C

C. 2

Choice C is correct. The solutions to a system of two equations correspond to points where the graphs of the equations intersect. The given graphs intersect at 2 points; therefore, the system has 2 solutions.

Choice A is incorrect because the graphs intersect. Choice B is incorrect because the graphs intersect more than once. Choice D is incorrect. It’s not possible for the graph of a quadratic equation and the graph of a linear equation to intersect more than twice.

Q3

A

A. $x^3 + 10x$

Choice A is correct. Applying the distributive property, the given expression can be written as $7x^3 + 7x - 6x^3 + 3x$. Grouping like terms in this expression yields $7x^3 - 6x^3 + 7x + 3x$. Combining like terms in this expression yields $x^3 + 10x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. 35

Choice B is correct. It's given that the table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. It's also given that when $x = 3$ the corresponding value of y is a, and a is a constant. Substituting 3 for x and a for y in the given equation yields $a = 42^3 + 3$, or $a = 35$. Therefore, the value of a is 35.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

D

D. $C = 19 - \frac{P}{N}$

Choice D is correct. It's given that the values of P, N, and C are positive. Therefore, dividing each side of the given equation by N yields $\frac{P}{N} = 19 - C$. Subtracting 19 from each side of this equation yields $\frac{P}{N} - 19 = -C$. Dividing each side of this equation by -1 yields $19 - \frac{P}{N} = C$, or $C = 19 - \frac{P}{N}$.

Choice A is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

Choice B is incorrect. This equation is equivalent to $P = 19 - NC$, not $P = N19 - C$.

Choice C is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

Q6

B

B. xy^2

Choice B is correct. One of the properties of radicals is $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$. Thus, the given expression can be rewritten as $\sqrt[3]{x^3} \cdot \sqrt[3]{y^6}$. Simplifying by taking the cube root of each part gives $x^1 \cdot y^2$, or xy^2.

Choices A, C, and D are incorrect and may be the result of incorrect application of the properties of exponents and radicals.

Q7

B

B. 18

Choice B is correct. The area, A, of a triangle is given by the formula $A = \frac{1}{2}bh$, where b represents the length of the base of the triangle and h represents its height. It's given that the area of a triangle is 270 square centimeters and that the length of the base of this triangle is 12 centimeters greater than the height of the triangle. Let x represent the height, in centimeters, of the triangle. It follows that the length of the base of the triangle can be expressed as $x + 12$. Substituting 270 for A, x for h, and $x + 12$ for b in the formula $A = \frac{1}{2}bh$ yields $270 = \frac{1}{2}\left(x + 12\right)\left(x\right)$, or $270 = \frac{1}{2}x\left(x + 12\right)$. Multiplying both sides of this equation by 2 yields $540 = x(x + 12)$. Applying the distributive property on the right-hand side of this equation yields $540 = x^2 + 12x$. Subtracting 540 from both sides of this equation yields $0 = x^2 + 12x - 540$. In factored form, this equation is equivalent to $(x + 30)(x - 18) = 0$. Applying the zero product property, it follows that $x + 30 = 0$ or $x - 18 = 0$. Subtracting 30 from both sides of the equation $x + 30 = 0$ yields $x = -30$. Adding 18 to both sides of the equation $x - 18 = 0$ yields $x = 18$. Since x represents the height of the triangle, it must be positive. Therefore, the height, in centimeters, of the triangle is 18.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

5

The correct answer is 5. The given quadratic equation can be rewritten in factored form as $(x - 8)(x + 3) = 0$. Based on the zero product property, it follows that $x - 8 = 0$ or $x + 3 = 0$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. Therefore, the solutions to the given equation are 8 and -3. It follows that the sum of the solutions to the given equation is $8 + (-3)$, or 5.

Q9

B

B. $7x^3 + 10$

Choice B is correct. The given expression is equivalent to $8x^3 + 8 - x^3 - -2$, or $8x^3 + 8 - x^3 + 2$. Combining like terms in this expression yields $7x^3 + 10$.

Choice A is incorrect. This expression is equivalent to $8x^3 + 8 - 2$, not $8x^3 + 8 - x^3 - 2$.

Choice C is incorrect. This expression is equivalent to $8x^3 + 8 - -2$, not $8x^3 + 8 - x^3 - 2$.

Choice D is incorrect. This expression is equivalent to $8x^3 + 8 - x^3 + 2$, not $8x^3 + 8 - x^3 - 2$.

Q10

A

A. If the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176$ ft^2.

Choice A is correct. The function f gives the area of the rectangle, in ft^2, if its width is w ft. Since the value of $f14$ is the value of fw if $w = 14$, it follows that $f14 = 1{,}176$ means that fw is $1{,}176$ if $w = 14$. In the given context, this means that if the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176$ ft^2.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from interpreting fw as the width, in ft, of the rectangle if its area is w ft^2, rather than as the area, in ft^2, of the rectangle if its width is w ft.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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